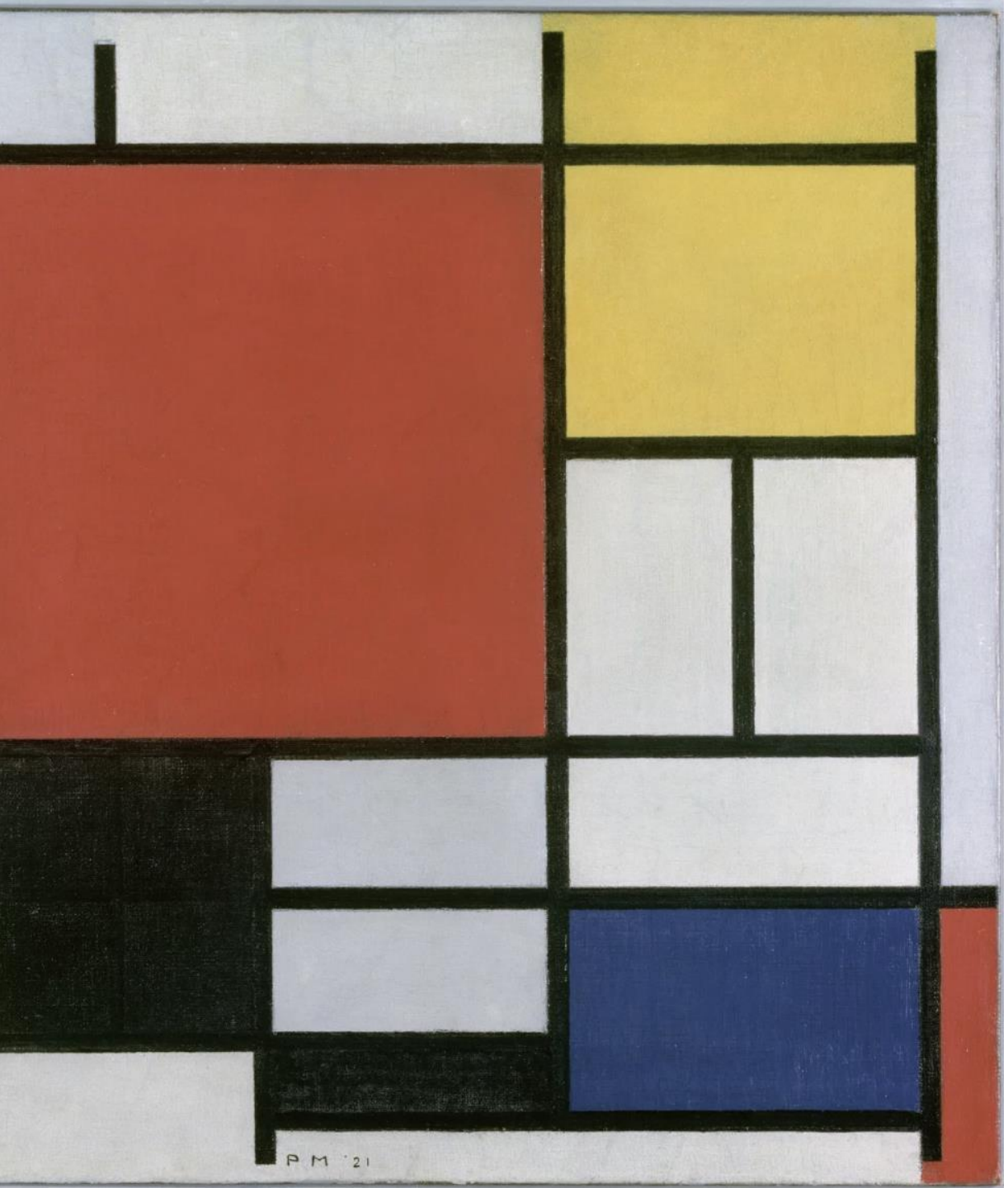


Mondrian Collage

20231005 Siyun Kim

Preload & Setup

```
function preload() {
  mon = loadImage('mondrian.png');
  for (let i = 0; i < MAX_IMG; i++) {
    images[i] = loadImage(`images/image${i + 1}.jpg`);
  }

}
```

1. Load images

```
  for (let j = 0; j < MAX_IMG; j++) {
    images[j].resize(imageWidth, imageHeight);
  }
```

2. Resize images

ImgSeparate

```javascript
function imgSeparate(img) {
  img.loadPixels();

  let rTotal = 0;
  let gTotal = 0;
  let bTotal = 0;
  let pixelCount = img.pixels.length / 4;

  for (let i = 0; i < img.pixels.length; i += 4) {
    let r = img.pixels[i];
    let g = img.pixels[i + 1];
    let b = img.pixels[i + 2];

    rTotal += r;
    gTotal += g;
    bTotal += b;
  }

  let rAvg = rTotal / pixelCount;
  let gAvg = gTotal / pixelCount;
  let bAvg = bTotal / pixelCount;
```

```javascript
  // RGB to HSB
  let avgColor = color(rAvg, gAvg, bAvg);

  colorMode(HSB);
  let h = hue(avgColor);

  // red range
  if (h >= 0 && h < 30) {
    reds.push(img);
  }
  // yellow range
  else if (h >= 30 && h < 90) {
    yellows.push(img);
  }
  // blue range
  else if (h >= 180 && h < 270) {
    blues.push(img);
  }

  colorMode(RGB);
}
```

3. Separate images
   → which color is its main color?

Draw

4. Draw images on appropriate location based on their main color

```
for (let r = 0; r < 28; r++) {
  if (r < 25) {
    x = 61 + imageWidth * (r % 5);
    y = 65 + imageHeight * floor(r / 5);
    image(reds[r], x, y);
  }
  else {
    x = 467;
    y = 378 + imageHeight * (r - 25);
    image(reds[r], x, y);
  }
}
```

```
for (let b = 0; b < 9; b++) {
  x = 312 + imageWidth * (b % 3);
  y = 315 + imageHeight * floor(b / 3);
  image(blues[b], x, y);
}
```

```
for (let yy = 0; yy < 15; yy++) {
  if (yy < 3) {
    x = 11;
    y = 378 + imageHeight * yy;
    image(yellows[yy], x, y);
  }
  else {
    x = 312 + imageWidth * (yy % 3);
    y = 0 + imageHeight * floor((yy - 3) / 3);
    image(yellows[yy], x, y);
  }
}
```

Draw

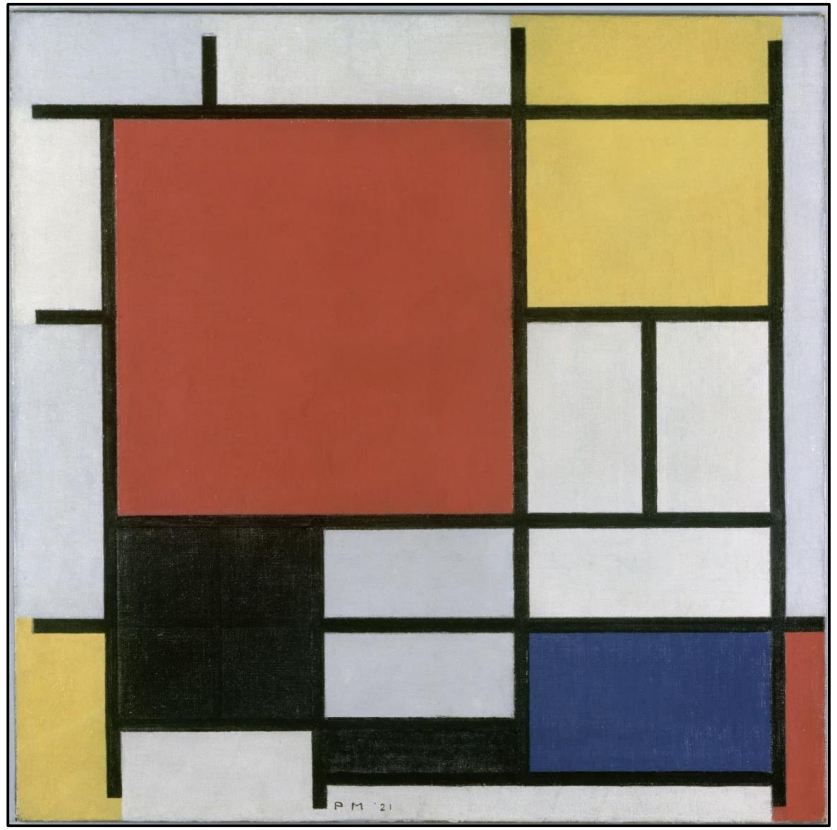
```
let x_left = 15;
let x_right = 465;

let y_head = 20;
let y_tail = 483;

stroke(0);
strokeWeight(8);
// draw vertical lines
line(61, 65, 61, y_tail);
line(122, y_head, 122, 65);
line(190, 315, 190, y_tail);
line(312, y_head, 312, 465);
line(390, 200, 390, 314);
line(x_right, y_head, x_right, y_tail);
// draw horizontal lines
line(x_left, 65, x_right, 65);
line(x_left, 200, 61, 200);
line(312, 200, x_right, 200);
line(61, 314, x_right, 314);
line(x_left, 378, width, 378);
line(61, 440, 310, 440);
line(190, 465, x_right, 465);
```

5. Draw other elements in
   [Composition of red blue and yellow]

Reference & Screenshot



https://editor.p5js.org/tldbs37/sketches/EeDwgf84v